06-05 第一篇

How Plants Defend Themselves

Plants cannot run away from danger. When an insect wants to eat their leaves or an animal steps on their stems, they have no legs to escape. [1] Yet plants are far from helpless—they have developed a wide range of clever ways to protect themselves over millions of years of evolution. From sharp thorns to bitter chemicals, plants use both physical and chemical weapons to survive.

Physical defences are the first line of protection. [2] Thorns, spines, and prickles make it painful or difficult for animals to eat or touch the plant. Cacti（仙人掌） have sharp spines that keep desert animals away. Roses have thorns that protect their flowers. Some plants have thick, waxy（蜡质的） leaves that are hard to bite or digest. Others grow small hairs on their leaves that make it uncomfortable for insects to walk on. These structures have one goal: to make the plant less attractive as food.

Plants also fight back with chemistry. [3] When an insect starts eating a leaf, some plants release chemicals that taste bitter or smell bad, driving the pest away. Others produce toxins（毒素） that can make the attacker sick or even die. The hot taste of chilli peppers, for example, comes from a chemical called capsaicin（辣椒素）, which mammals find painful but birds can eat without harm. This helps chilli plants spread their seeds—birds eat the fruit and drop the seeds in new places, while mammals learn to avoid them.

Some plants have even more surprising strategies. [4] When under attack, certain plants can send chemical signals through the air to warn nearby plants of the same species. Those neighbouring plants then start producing defence chemicals before the insects arrive. Scientists have also found that some plants can attract beneficial insects—such as wasps that eat harmful caterpillars—by releasing special scents. It is almost as if plants can "talk" to each other and to their insect allies.

[5] Plants may seem passive, but they are skilled survivors. Their defence systems have evolved over millions of years and continue to amaze scientists today. The next time you admire a flower or avoid a thorny bush, remember—you are looking at a master of self-protection.

选项

A. In other words, plants have evolved many different strategies to stay safe from harm. B. That is because these structures act as barriers that discourage animals from feeding on the plant. C. In fact, plants can use chemicals to repel attackers and even to communicate with other plants. D. These advanced behaviours show that plants are

more intelligent than we once thought. E. In short, plants are experts at defending themselves, even without the ability to move. F. For example, the Venus flytrap catches insects to get nutrients from poor soil.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：引出植物防御的多样性。空前描述"Plants cannot run away from danger...have no legs to escape"，空后说明"Yet plants are far from helpless—they have developed a wide range of clever ways to protect themselves...From sharp thorns to bitter chemicals...physical and chemical weapons"。A选项"In other words, plants have evolved many different strategies to stay safe from harm"（换言之，植物进化出多种不同策略以保护自己免受伤害）中"In other words"承接空前困境与空后转折，"many different strategies"与空后"wide range of clever ways...thorns...chemicals"精准呼应，衔接顺畅。
- **[2] B**：
  - **线索**：物理防御的作用机制。空前提出"Physical defences are the first line of protection"，空后列举"Thorns, spines, prickles...painful or difficult for animals...Cacti...sharp spines...Roses have thorns...thick waxy leaves...small hairs...make the plant less attractive as food"。B选项"That is because these structures act as barriers that discourage animals from feeding on the plant"（那是因为这些结构充当屏障，阻止动物取食植物）中"That is because"解释物理防御的功能，"barriers...discourage...feeding"与空后"painful...difficult...keep away...protect...hard to bite...less attractive"完美对应，逻辑紧密。
- **[3] C**：
  - **线索**：化学防御的机制。空前提出"Plants also fight back with chemistry"，空后详述"When an insect starts eating a leaf...release chemicals that taste bitter or smell bad, driving the pest away...produce toxins...chilli peppers...capsaicin"。C选项"In fact, plants can use chemicals to repel attackers and even to communicate with other plants"（事实上，植物能用化学物质驱赶攻击者，甚至与其他植物交流）中"In fact"强化空前化学战，"repel attackers"与空后苦味、臭味、毒素、辣椒素等驱赶或毒杀攻击者的机制精准对应，衔接自然。
- **[4] D**：

- **线索**：植物的高级防御行为。空前提出"Some plants have even more surprising strategies"，空后描述"send chemical signals...warn nearby plants...neighbouring plants...produce defence chemicals...attract beneficial insects...wasps that eat caterpillars...release special scents...plants can 'talk' to each other and to insect allies"。D选项"These advanced behaviours show that plants are more intelligent than we once thought"（这些高级行为表明植物比我们曾经认为的更聪明）中"These advanced behaviours"指代空前"more surprising strategies"，"more intelligent"与空后化学信号、预警、吸引益虫、"talk"等复杂行为完美对应，衔接得当。

- **[5] E**：
  - **线索**：全文收束与升华。空[5]位于结尾段段首，空后总结"Plants may seem passive, but they are skilled survivors...defence systems have evolved over millions of years...master of self-protection"。E选项"In short, plants are experts at defending themselves, even without the ability to move"（简言之，植物是自我防御的专家，即便无法移动）用"In short"总结全文，"experts at defending...without ability to move"与空后"skilled survivors...master of self-protection"及全文"cannot run away"形成呼应，收束精准。

- **干扰项分析**：
  - **F. For example, the Venus flytrap catches insects to get nutrients from poor soil.**：涉及捕蝇草捕虫获取营养，易被误填[3]或[4]。但F以"For example"引出捕蝇草这一主动捕食的特例，而空[3]前后文讨论的是释放化学物质驱赶或毒害攻击者的被动化学防御，捕蝇草属于主动捕食获取营养，与"bitter, toxins, capsaicin"等化学防御机制不同；空[4]讨论的是植物间通讯与吸引益虫，F的"catches insects"为捕食行为，非通讯或协作。F属"主题相关但防御类型错位"型干扰。